AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM358
Machine Learning & Artificial intelligence

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 6/Dec. /2025

Number of Exam Pages: ... (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 ..... algorithms that try to find the best solution and to prove that it is the best.

A exact
B optimization
C combinational optimization
D approximate

2 ..... is more concerned with obtaining the correct response to an input stimulus without 'looking inside the box' to see if parts of the mechanism can be associated with discrete real world objects.

A Symbolic
B Subsymbolic
C Algebric
D None of the above

3 The technical term for when completing one sub-plan undoes the achievements of another is .....

A planning
B Block world
C clobbering
D None of the above

4 ..... sensors measure range based on physical contact, and can be deployed only for sensing objects very close to the robot.

A All of the above
B Range finders
C Sonar sensors
D Tactile sensors

5 ..... knowledge deals with factoid questions

A Declarative
B Instructional
C All of the above
D Procedural

6 In ..... the solution will be the best possible combination of elements of some given set, such as in the scheduling of jobs in a factory

A combinational optimization
B adaptation
C optimization
D selection

7 ..... just like DFS, except never go deeper than some depth d

A BFS
B Depth-limited
C Best first
D A*

8 ..... Means "Is the algorithm guaranteed to find a solution when there is one?"

A Space complexity
B Completeness
C Optimality
D Time complexity

9 Chess game is an example of .....games.

A stochastic
B deterministic
C random
D None of the above

10 insect movement is an example of .....

A interaction with other agents
B interaction with the environment
C internal interaction
D None of the above

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

a. Describe the two main types of search in the context of AI and explain the key differences between them. [5 marks]

b. Given the following search tree for *the stone game(5,3)*, use expectiminimax algorithm to calculate the value of each node hence decide the winner for that game.

Assume the following switching probabilities for min player:

Action	p
K[1]	0.50
L[0]	0.50
W[1]	0.69
X[0]	0.31
H[2]	0.41
I[1]	0.14
J[0]	0.45
E[3]	0.29
F[2]	0.14
G[1]	0.57



Question 3: (15 marks)

a. Explain the process of agenda-based planning in STRIPS. How does the system handle goal conditions that are already true?

b. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

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s	A	B	B	B	C	C	D	D	D	D	E	E
t	B	C	D	E	F	G	H	I	J	K	L	M

Given the aforementioned tree, apply BFS to traverse the tree showing all agenda content and traced paths until reaching the goal state if exist assuming that the goal state is I

Question 4: (15 marks)

Having the following current state s for XO game=

O	E	X
E	E	O
O	X	X

a. Write the direct children states for s following the XO actions.
b. Mention one possible heuristic function for this game
c. Based on b, calculate the heuristic value for each child state obtained in a.

End of Questions

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